SAC-Assisted Hybrid Control for Quadrotor Path Following under Wind Disturbances

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***Abstract*—Quadrotor path-following performance can be degraded by wind disturbances, especially when a fixed-gain baseline controller cannot sufficiently compensate for residual tracking errors. This paper proposes a hybrid residual control scheme that augments a conventional P-PD-based quadrotor controller with a Soft Actor-Critic (SAC)-based auxiliary thrust controller. The baseline controller provides the nominal stabilizing action, while the SAC agent generates bounded auxiliary thrust commands at the rotor thrust level. A Huber-loss-based reward function is designed using position, velocity, attitude, and angular-rate error terms to train the residual policy under wind disturbance conditions. The proposed controller was evaluated in four simulation scenarios, including hovering, line-segment tracking, circular path following, and figure-eight path following. Under the main OU-type stochastic wind disturbance condition, the SAC-assisted controller reduced the three-dimensional position RMSE by 31.82%, 13.15%, 11.75%, and 26.29% in the four scenarios, respectively, compared with the baseline controller without reinforcement learning assistance. Compared with DDPG- and TD3-assisted controllers, SAC showed the most consistent improvement among the tested RL-based auxiliary controllers. Additional evaluations under multi-harmonic sinusoidal wind disturbance, reward-weight variations, thrust saturation constraints, and inference-time measurement further supported the robustness and feasibility of the proposed residual control structure. These results indicate that SAC-based residual reinforcement learning can improve quadrotor path-following performance under wind disturbances while preserving the stabilizing role of the conventional baseline controller.**

***Keywords*—Quadrotor Control, Hybrid Control, Soft Actor-Critic, Residual Reinforcement Learning, Path Following, Disturbance Rejection**

I. INTRODUCTION

Quadrotors have been widely used in various applications, including inspection, delivery, surveillance, and disaster response, because of their vertical takeoff and landing capability and high maneuverability. In urban unmanned aircraft and urban air mobility operations, multiple vehicles may operate within limited low-altitude airspace, and therefore safe operation requires compliance with predefined routes, airspace structures, and traffic management concepts [1], [2]. However, quadrotor flight is strongly affected by external disturbances such as wind gusts, as well as modeling uncertainties arising from nonlinear and coupled dynamics. These factors can degrade tracking accuracy when the vehicle follows predefined paths under time-varying disturbance conditions. Therefore, improving path-following performance under wind disturbances remains an important problem in quadrotor control.

Conventional PID- or P-PD-based controllers have been widely used as practical baseline controllers for quadrotor stabilization and tracking because of their simple structure, low computational cost, and ease of implementation [3]. However, when the vehicle is operated under wind disturbances and model uncertainties, the fixed-gain baseline controller alone may not sufficiently compensate for residual tracking errors along predefined paths. Therefore, an auxiliary control structure is needed to improve disturbance compensation while preserving the stabilizing role of the conventional controller.

To address this issue, this paper adopts a residual reinforcement learning framework in which a learned policy compensates for residual tracking errors while the conventional controller retains the primary stabilizing role [4]. Similar residual learning concepts have also been applied to cascaded PID-controlled quadcopters to improve wind resistance while preserving the baseline controller structure [5]. In the proposed structure, the P-PD-based baseline controller computes the nominal rotor thrusts, and a Soft Actor-Critic (SAC) agent generates bounded auxiliary thrust commands that are additively applied to each motor input. This configuration restricts the influence of the learning agent to residual compensation rather than allowing it to replace the baseline controller. In addition, the reward function is formulated using the Huber loss to reduce excessive sensitivity to occasional large errors caused by disturbances while maintaining sensitivity to small tracking errors. The reward includes position, velocity, attitude, and angular-rate error terms so that

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both path-following performance and attitude stabilization are considered during learning.

The main contributions of this paper are summarized as follows. First, a hybrid residual control structure is designed by augmenting a conventional P-PD-based quadrotor controller with a SAC-based auxiliary thrust controller at the rotor thrust level. Second, a Huber-loss-based reward function is formulated for residual policy learning by incorporating position, velocity, attitude, and angular-rate error terms under wind disturbance conditions. Third, the proposed SAC-assisted controller is evaluated in four path-following scenarios and compared with the No Agent baseline as well as DDPG- and TD3-assisted auxiliary controllers. In addition, a reward-weight sensitivity analysis is conducted to examine the influence of the vector-following and attitude-stabilization weights on tracking performance. Fourth, additional evaluations are performed under an unseen multi-harmonic sinusoidal wind disturbance, thrust saturation constraints, and policy inference-time measurement to examine the robustness and feasibility of the proposed residual control structure.

II. Proposed Hybrid Residual Control Method

A. Baseline P-PD Control Structure

The baseline controller is implemented as a cascaded control structure that converts the reference velocity vector into individual rotor thrust commands. The reference vector generator provides the desired velocity vector according to the current position and the predefined path. The translational controller then computes the velocity tracking error as

$$\mathbf{e}_v = \mathbf{v}_{\text{ref}} - \mathbf{v}, \tag{1}$$

where $\mathbf{v}_{\text{ref}} = \left[v_{x,\text{ref}}, v_{y,\text{ref}}, v_{z,\text{ref}}\right]^T$ is the reference velocity vector and $\mathbf{v} = \left[v_x, v_y, v_z\right]^T$ is the current translational velocity. The desired translational acceleration is generated by a proportional controller as

$$\mathbf{a}_{\text{des}} = \mathbf{K}_v \mathbf{e}_v, \tag{2}$$

where $\mathbf{K}_v$ is the translational proportional gain matrix.

The desired translational acceleration is converted into the desired roll and pitch angles and the total thrust command. The attitude controller then computes virtual torque commands using proportional and derivative feedback of the attitude and angular-rate errors. The resulting total thrust and virtual torque commands are distributed to the four rotor thrusts through the thrust mixer. The baseline motor thrust vector is denoted as

$$\mathbf{T}_b = \left[T_{b,1}, T_{b,2}, T_{b,3}, T_{b,4}\right]^T. \tag{3}$$

This baseline controller provides the nominal stabilizing control action and is retained in both the baseline-only and proposed control cases. In the proposed method, the reinforcement learning agent does not replace this controller; instead, it generates auxiliary thrust commands that are added to $\mathbf{T}_b$, as described in the next subsection.

B. SAC-Based Auxiliary Thrust Control

SAC is an off-policy actor-critic algorithm based on the maximum entropy reinforcement learning framework. Unlike standard reinforcement learning objectives that maximize only the expected return, SAC maximizes a trade-off between the expected return and the policy entropy. The objective can be written as

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(s_t,a_t)\sim\rho_\pi}[r(s_t,a_t) + \alpha\mathcal{H}(\pi(\cdot\,|s_t))] \tag{4}$$

where s_t is the state, a_t is the action, $r(s_t,a_t)$ is the reward, $\mathcal{H}(\pi(\cdot\,|\,s_t))$ is the entropy of the policy at state s_t, and α is the temperature parameter that determines the relative importance of the entropy term. This entropy-regularized objective encourages the policy to maintain stochasticity during learning, which can improve exploration and reduce premature convergence to overly deterministic actions [6].

This property is useful for quadrotor residual control under wind disturbances because the disturbance sequence is time-varying and cannot be fully compensated for by the fixed-gain baseline controller. In this study, SAC is not used to replace the conventional controller. Instead, it is used as an auxiliary controller that learns residual thrust corrections for the four rotor inputs. By maintaining the baseline P-PD controller, the proposed structure preserves the nominal stabilizing action, while the SAC agent is trained to compensate for residual tracking errors caused by wind disturbances and modeling uncertainties.

The observation vector of the SAC agent is defined as

$$\mathbf{o} = \left[e_x, e_y, e_z, e_{v_x}, e_{v_y}, e_{v_z}, \phi, \theta, \psi, p, q, r\right]^T. \tag{5}$$

The observation vector $\mathbf{o}$ consists of the position error, translational velocity error, Euler angles, and body angular rates of the quadrotor. Here, $\left[e_x, e_y, e_z\right]^T = \mathbf{p}_{\text{ref}} - \mathbf{p}$ denotes the position-error vector, and $\left[e_{v_x}, e_{v_y}, e_{v_z}\right]^T = \mathbf{v}_{\text{ref}} - \mathbf{v}$ denotes the translational velocity-error vector. The variables ϕ, θ, and ψ denote the roll, pitch, and yaw angles; and p, q, and r denote the body angular rates.

The final motor thrust vector is obtained by adding the SAC-generated auxiliary thrust command to the baseline motor thrust vector as

$$\mathbf{T} = \text{sat}(\mathbf{T}_b + \mathbf{\Delta T}, \mathbf{0}, \mathbf{T}_{\max}), \tag{6}$$

where $\mathbf{T}_b$ is the baseline motor thrust vector computed in the previous subsection, $\mathbf{T}_{\max}$ denotes the prescribed upper bound of the final motor thrust vector, and $\text{sat}(\cdot)$ denotes the saturation function that limits each motor thrust within its allowable range.

The auxiliary thrust command is defined as

$$\mathbf{\Delta T} = [\Delta T_1, \Delta T_2, \Delta T_3, \Delta T_4]^T, \tag{7}$$

where each component ΔT_i is applied to the corresponding rotor. To prevent excessive residual compensation, each auxiliary thrust command was bounded by

$$\Delta T_i \in [-0.02T_{\text{lim}}, 0.02T_{\text{lim}}],\ i = 1, \cdots, 4, \tag{8}$$

where T_{lim} denotes the baseline total thrust limit. In this study, $T_{\text{lim}} = 1.5mg$, and therefore the maximum auxiliary thrust magnitude applied to each motor was $0.02T_{\text{lim}}$. This additional

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bounded thrust margin was introduced so that the residual controller was not immediately constrained by the baseline thrust headroom. Therefore, $T_{\lim}$ should be interpreted as the baseline thrust limit, whereas $\mathbf{T}_{\max}$ defines the allowable final motor thrust range after residual compensation. The numerical values of $T_{\lim}$, the per-motor upper bound $T_{\max,i}$, and the resulting saturation rates are provided and analyzed in Section IV-C, where $T_{\max,i}$ denotes the i-th component of $\mathbf{T}_{\max}$.

C. Huber-Loss-Based Reward Function

To train the SAC-based residual policy, the reward function is designed to reflect both path-following performance and attitude stabilization. Under wind disturbance conditions, occasional large errors may occur due to transient responses and time-varying external disturbances. Therefore, Huber loss is used to reduce excessive sensitivity to large errors while maintaining sensitivity to small tracking errors [7]. The Huber loss is defined as

$$L_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \le \delta, \\ \delta\left(|e| - \frac{1}{2}\delta\right), & |e| > \delta, \end{cases} \tag{9}$$

where e is an error variable and δ is the threshold parameter. The position error is defined using the reference position $\mathbf{p}_{\text{ref}}$ and the current position $\mathbf{p}$ as

$$\mathbf{e}_p = \mathbf{p}_{\text{ref}} - \mathbf{p}. \tag{10}$$

For velocity tracking, both the speed magnitude error and the velocity direction error are considered. The speed magnitude error is defined as

$$e_{v,\text{mag}} = |\|\mathbf{v}_{\text{ref}}\| - \|\mathbf{v}\|| \tag{11}$$

And the velocity direction error is defined as

$$e_{v,\text{dir}} = w_d(1 - \hat{\mathbf{v}}_{\text{ref}}^T\hat{\mathbf{v}}) \tag{12}$$

where $\hat{\mathbf{v}}_{\text{ref}}$ and $\hat{\mathbf{v}}$ denote the normalized reference and current velocity vectors, respectively. The coefficient w_d reduces the effect of the direction error when either velocity magnitude is close to zero.

The attitude and angular-rate errors are defined as

$$\mathbf{e}_\eta = \boldsymbol{\eta}_{\text{ref}} - \boldsymbol{\eta}, \ \mathbf{e}_\omega = \boldsymbol{\omega}_{\text{ref}} - \boldsymbol{\omega}, \tag{13}$$

where $\boldsymbol{\eta} = [\phi, \theta, \psi]^T$ and $\boldsymbol{\omega} = [p, q, r]^T$. In this study, the reference attitude and angular-rate values are set to zero for stabilization.

The Huber-loss-based vector-following loss and attitude-stabilization loss are then defined as

$$L_{\text{vec}} = \sum_{i\in\{x,y,z\}} L_{\delta_p}(e_{p,i}) + L_{\delta_m}(e_{v,\text{mag}}) + L_{\delta_d}(e_{v,\text{dir}}), \tag{14}$$

$$L_{\text{att}} = \sum_{i\in\{\phi,\theta,\psi\}} L_{\delta_\eta}(e_{\eta,i}) + \sum_{i\in\{p,q,r\}} L_{\delta_\omega}(e_{\omega,i}). \tag{15}$$

Finally, the reward is defined as the negative weighted sum of these two loss terms:

$$r = -(\lambda_v L_{\text{vec}} + \lambda_a L_{\text{att}}), \tag{16}$$

where λ_v and λ_a are weighting coefficients for the vector-following and attitude-stabilization terms, respectively. Through this formulation, the SAC agent is trained to reduce position and velocity-tracking errors while maintaining attitude stability under wind disturbance conditions.

III. Simulation Setup

A. Simulation Environment and Wind Disturbance

The proposed control structure was evaluated using a MATLAB/Simulink-based quadrotor simulation environment. The simulation model consists of the reference vector generator, the baseline P-PD controller, the thrust mixer, the SAC-based auxiliary thrust controller, the quadrotor dynamics, and the wind disturbance model. The baseline controller and the SAC agent are connected at the rotor thrust level so that the auxiliary thrust commands generated by the SAC agent are added to the nominal motor thrusts computed by the baseline controller.

The quadrotor was modeled as a six-degree-of-freedom rigid-body system, following the commonly used Newton-Euler formulation for quadrotor dynamics [8]. The translational and rotational dynamics are represented as

$$m\ddot{\mathbf{p}} = \mathbf{R}\mathbf{f}_T + \mathbf{f}_g + \mathbf{f}_w, \tag{17}$$

$$\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega}, \tag{18}$$

where $\mathbf{p} = [x, y, z]^T$ is the position vector, m is the mass, $\mathbf{R}$ is the rotation matrix from the body frame to the inertial frame, $\mathbf{f}_T = [0,0,U_1]^T$ is the thrust force vector, $\mathbf{f}_g = [0,0,-mg]^T$ is the gravity force vector, and $\mathbf{f}_w$ is the wind disturbance force. In the rotational dynamics, $\mathbf{J} = \text{diag}(I_x, I_y, I_z)$ is the inertia matrix, $\boldsymbol{\omega} = [p, q, r]^T$ is the body angular-rate vector, and $\boldsymbol{\tau} = [U_2, U_3, U_4]^T$ is the torque input vector.

The main physical parameters used in the simulation are summarized in Table 1. These parameters include the vehicle mass, arm length, inertia components, yaw torque coefficient, and gravitational acceleration.

Table 1. Quadrotor Simulation Parameters

Parameter	Value	Unit
Mass m	4.0	kg
Arm length l	0.32	m
Moment of inertia I_x	0.07	kg·m²
Moment of inertia I_y	0.07	kg·m²
Moment of inertia I_z	0.12	kg·m²
Yaw torque coefficient k_ψ	0.05	N·m/N
Gravitational acceleration g	9.81	m/s²

Wind disturbance was modeled as a time-correlated stochastic disturbance using an Ornstein-Uhlenbeck process, which has been used for stochastic atmospheric turbulence and wind-profile generation because of its Markov property and exponential covariance structure [9]. The generated disturbance was applied to the translational dynamics as equivalent external force components along the inertial x, y, and z axes. Through this configuration, the proposed controller was evaluated under wind disturbance conditions that vary continuously over time.

B. Evaluation Scenarios and Metrics

The proposed controller was evaluated in four representative path-following scenarios, as summarized in Table 2. The scenarios include fixed-point hovering, line-segment tracking, circular path following, and figure-eight path following. These cases were selected to evaluate the controller

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under different reference-vector generation conditions, ranging from stationary hovering to continuously varying curved paths.

Table 2. EVALUATION SCENARIOS

Case	Scenario	Description
Case 1	Hovering	Hovering at a fixed reference position
Case 2	Line segment	Tracking a predefined line-segment path
Case 3	Circular path	Tracking a circular path with a fixed radius
Case 4	Figure-eight path	Tracking a figure-eight path generated by a Lissajous curve

The SAC policy was trained under the hovering condition using the OU-type stochastic wind disturbance. However, the policy input was defined using tracking-error-related states rather than trajectory-specific indices, while the baseline P-PD controller remained responsible for the nominal trajectory-tracking task. Therefore, the SAC agent was not intended to learn a trajectory-specific path-following policy, but to act as a bounded residual compensator for the same error-regulation problem. From this perspective, Cases 2–4 were used to evaluate the applicability of the hover-trained residual policy under different reference-motion conditions without trajectory-specific retraining. Nevertheless, the degree of improvement can differ depending on the path geometry and disturbance-response characteristics. The additional sinusoidal wind disturbance was used only for post-training robustness assessment.

The performance of the proposed controller was first evaluated by comparing the baseline controller without reinforcement learning assistance, referred to as No Agent, and the proposed SAC-assisted hybrid controller. Additional RL-based auxiliary controllers, including DDPG and TD3, were also evaluated for algorithmic comparison in the main disturbance condition. To ensure a fair comparison, all controllers were evaluated under the same wind disturbance seed sequences. In the evaluation stage, 30 independent disturbance seeds were used for each scenario, and the simulation results were collected for each control case.

The primary evaluation metric was the three-dimensional position root mean square error (RMSE), which is defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\|\mathbf{p}_{\mathrm{ref}}(k) - \mathbf{p}(k)\|^2}, \qquad (19)$$

where N is the number of time samples, $\mathbf{p}_{\mathrm{ref}}(k)$ is the reference position, and $\mathbf{p}(k)$ is the current position at the k-th time step. For each simulation run, RMSE was calculated over the simulation time and compared across the four scenarios. This metric was selected because the main objective of this study is to improve path-following accuracy under wind disturbance conditions.

IV. RESULTS AND DISCUSSION

A. *Quantitative Comparison*

The quantitative path-following performance was evaluated using the three-dimensional position RMSE defined in (19). Table 3 summarizes the RMSE values of the baseline controller without reinforcement learning assistance, referred to as No Agent, and the RL-assisted controllers based on DDPG, TD3, and SAC. The improvement rate was computed relative to the No Agent case. DDPG and TD3 were selected as representative off-policy actor–critic algorithms for continuous control, enabling a direct comparison with SAC under the same auxiliary-thrust control framework.

Table 3. 3D POSITION RMSE COMPARISON

Method	Metric	Case 1	Case 2	Case 3	Case 4
No Agent	RMSE (m)	0.0733	0.1234	0.1678	0.2194
DDPG	RMSE (m)	0.0615	0.1880	0.2067	0.1978
	Improvement (%)	16.04	-52.30	-23.18	9.87
TD3	RMSE (m)	0.0654	0.1481	0.2002	0.1788
	Improvement (%)	10.75	-20.00	-19.33	18.50
SAC	RMSE (m)	0.0500	0.1072	0.1481	0.1618
	Improvement (%)	31.82	13.15	11.75	26.29

Table 3 compares the three-dimensional position RMSE of the No Agent baseline with those of the DDPG-assisted, TD3-assisted, and SAC-assisted controllers. The SAC-assisted controller achieved positive improvement rates over the No Agent baseline in all four evaluation cases. Specifically, the 3D position RMSE was reduced from 0.0733 m to 0.0500 m in Case 1, corresponding to a 31.82% improvement. In Cases 2, 3, and 4, the SAC-assisted controller reduced the RMSE by 13.15%, 11.75%, and 26.29%, respectively. In contrast, DDPG and TD3 improved the 3D position RMSE over the No Agent baseline in Cases 1 and 4 but degraded the performance in Cases 2 and 3. For DDPG, the RMSE increased by 52.30% and 23.18% in Cases 2 and 3, respectively, while TD3 showed increases of 20.00% and 19.33% in the same cases. These results indicate that SAC provided the most consistent improvement in 3D position tracking among the tested RL-based auxiliary controllers under the evaluated disturbance and trajectory conditions.

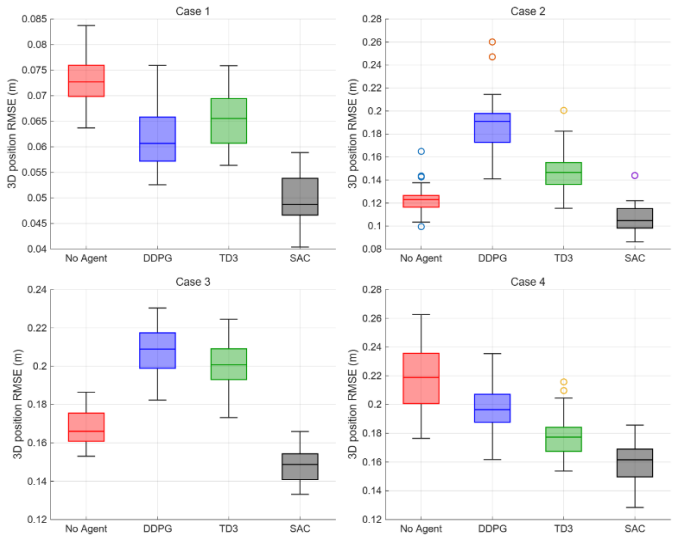


Figure 1. Distribution of 3D position RMSE for No Agent, DDPG, TD3, and SAC across four evaluation cases over 30 runs.

The boxplot results in Figure 1 further illustrate the run-to-run distribution of the 3D position RMSE for the No Agent, DDPG-, TD3-, and SAC-assisted controllers. The SAC-assisted controller exhibited the lowest median RMSE in all four cases and maintained a relatively compact error distribution. In contrast, DDPG and TD3 showed higher RMSE distributions than the No Agent baseline in Cases 2 and 3, which is consistent

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with the negative improvement rates reported in Table 3. These results indicate that SAC provided the most consistent 3D position-tracking performance among the tested RL-based auxiliary controllers under the evaluated wind disturbance conditions.

To further examine the sensitivity of the proposed SAC-assisted controller to reward-weight selection, a one-factor-at-a-time ablation study was conducted for λ_a and λ_v. Since the SAC agent was implemented using MATLAB Reinforcement Learning Toolbox, the entropy-temperature parameter α was handled internally and was not manually varied. Each reward weight was varied by $0.5\times$ and $2.0\times$ from its nominal value, while the other weight was fixed at its nominal value. Table 4 summarizes the 3D position RMSE improvement rates over the No Agent baseline for the four evaluation cases.

Table 4. REWARD-WEIGHT SENSITIVITY ANALYSIS

Setting	λ_a	λ_v	Case 1	Case 2	Case 3	Case 4
Nominal	1.0	1.0	31.82	13.15	11.75	26.29
A-high	2.0	1.0	26.95	8.59	6.16	25.80
A-low	0.5	1.0	34.45	13.68	14.19	23.48
V-low	1.0	0.5	17.97	-1.09	-2.44	15.98
V-high	1.0	2.0	32.42	14.81	11.23	21.10

The results show that the SAC-assisted controller maintained positive improvement over the No Agent baseline in most tested settings. The performance was more sensitive to λ_v than to λ_a. In particular, reducing λ_v to $0.5\lambda_{v,0}$ resulted in negative improvement rates in Cases 2 and 3, whereas varying λ_a within the tested range maintained positive improvement in all four cases. These results indicate that the velocity-related reward weight plays a more critical role in maintaining path-following performance than the attitude-related reward weight under the tested conditions.

B. Additional Disturbance Evaluation

To further evaluate the disturbance robustness of the proposed controller, an additional multi-harmonic sinusoidal wind disturbance was applied. The sinusoidal disturbance was selected to test a structured periodic disturbance pattern that differs from the temporally correlated stochastic OU process used during training and in the main evaluation. This unseen disturbance condition was used to examine whether the learned residual compensation effect could be maintained under a disturbance pattern different from the training wind model.

The additional disturbance was generated as a multi-harmonic sinusoidal force along the x, y, and z axes. The fundamental frequency was set to f_0=0.20 Hz, and the third-, fifth-, and seventh-harmonic components were added, corresponding to 0.60, 1.00, and 1.40 Hz, respectively. After applying the axis-wise disturbance gains used in the Simulink model, the harmonic amplitudes were $[6.0,3.0,2.0,1.0]$ N for the x and y axes and $[1.8,0.9,0.6,0.3]$ N for the z axis. The phases of the harmonic components were generated deterministically from the evaluation seed in the range $[0, 2\pi)$. For each case, 30 evaluation runs were conducted using different random seeds to generate distinct but reproducible phase combinations.

Table 5. SINUSOIDAL DISTURBANCE EVALUATION

Method	Case 1	Case 2	Case 3	Case 4
No Agent RMSE (m)	0.1726	0.2475	0.2222	0.3652
SAC RMSE (m)	0.1531	0.2162	0.1949	0.2884
Improvement (%)	11.31	12.66	12.29	21.04

As shown in Table 5, the SAC-assisted controller reduced the 3D position RMSE in all four cases under the additional sinusoidal wind disturbance. The 3D position RMSE improvements were 11.31%, 12.66%, 12.29%, and 21.04% in Cases 1–4, respectively. These results indicate that although the SAC policy was trained under the OU-type stochastic wind condition, the learned residual compensation effect was maintained under an unseen periodic multi-harmonic disturbance condition.

C. Thrust Saturation Comparison among RL Algorithms

To examine whether the RL-based auxiliary thrust remained within the prescribed motor-thrust constraints, the thrust saturation behavior was analyzed. The baseline thrust limit of $1.5mg$ was applied only to the baseline controller output, not to the final thrust command after adding the auxiliary thrust. This setting was adopted to provide additional thrust headroom for the residual controller so that the auxiliary policy was not immediately constrained by the baseline controller limit. Therefore, the final thrust command can exceed the baseline thrust limit of $1.5mg$, but only within the predefined auxiliary-thrust bound.

In this study, the baseline total thrust limit was defined as $T_{\mathrm{lim}} = 1.5mg = 58.86\,\mathrm{N}$, corresponding to a baseline per-motor thrust limit of $14.715\,\mathrm{N}$. The RL-generated auxiliary thrust was bounded by $\pm 0.02T_{\mathrm{lim}}$ for each motor, resulting in a per-motor auxiliary-thrust limit of ± 1.1772 N. Consequently, the maximum allowable final motor thrust for each motor was set to $T_{\mathrm{max},i} = 15.8922\,\mathrm{N}$, where $T_{\mathrm{max},i}$ denotes the i-th component of $\mathbf{T}_{\mathrm{max}}$. The following analysis evaluates how frequently the auxiliary thrust reached its saturation bound and how long the final motor thrust remained at the prescribed saturation limit.

Table 6. THRUST SATURATION COMPARISON

Case	Metric	DDPG	TD3	SAC
1	Aux. sat. avg. (%)	4.53	0.61	0.00
	Total sat. time avg. (%)	0.00	0.00	0.00
2	Aux. sat. avg. (%)	18.02	6.00	0.05
	Total sat. time avg. (%)	0.39	0.22	0.05
3	Aux. sat. avg. (%)	35.53	12.42	0.04
	Total sat. time avg. (%)	0.08	0.08	0.02
4	Aux. sat. avg. (%)	50.47	24.39	0.03
	Total sat. time avg. (%)	0.34	0.30	0.01

As shown in Table 6, SAC exhibited the lowest auxiliary-thrust saturation rate among the tested RL algorithms in all cases. The average auxiliary-thrust saturation rate of SAC remained below 0.06%, whereas DDPG and TD3 reached 50.47% and 24.39%, respectively, in Case 4. The peak total thrust did not exceed the prescribed maximum allowable thrust in any of the evaluated cases, and the average total-thrust saturation time of SAC remained below 0.06% across all cases. These results indicate that the SAC-assisted controller improved the position-tracking performance without relying on sustained thrust saturation. Therefore, under the same auxiliary-thrust constraint, SAC provided the most stable and efficient

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use of the residual thrust command compared with DDPG and TD3.

D. Inference Time and Real-Time Feasibility

Table 7. SAC POLICY INFERENCE-TIME MEASUREMENT

Metric	Value
Warm-up calls	100
Repeated evaluations	10,000
Mean inference time	0.998 ms
95th percentile inference time	1.428 ms
Control period T_s	50 ms
Mean / T_s	1.99%
95th percentile / T_s	2.85%

As shown in Table 7, to evaluate the real-time feasibility of the proposed auxiliary controller, the inference time of the trained SAC policy was measured over 10,000 repeated forward evaluations after 100 warm-up calls. The mean SAC inference time was 0.998 ms, and the 95th percentile was 1.428 ms. Since the control period was T_s=50 ms, the mean and 95th-percentile inference times correspond to 1.99% and 2.85% of the control period, respectively. These results indicate that the SAC inference cost is sufficiently small relative to the simulation control period in the tested CPU-based MATLAB environment. However, validation on embedded flight hardware remains future work.

The inference-time measurement was conducted in a CPU-based MATLAB environment on a Windows 11 desktop computer equipped with a 12th Gen Intel Core i5-12500 CPU and 32 GB RAM, using MATLAB R2025b, Simulink R2025b, and Reinforcement Learning Toolbox R2025b.

V. CONCLUSION

This paper proposed a SAC-assisted hybrid residual control structure for quadrotor path following under wind disturbance conditions. In the proposed method, a conventional P-PD-based controller provides the nominal stabilizing control action, while the SAC agent generates bounded auxiliary thrust commands at the rotor thrust level. The learning agent therefore acts as a residual compensator rather than replacing the baseline controller. A Huber-loss-based reward function was designed using position, velocity, attitude, and angular-rate error terms to train the residual policy under wind disturbance conditions.

The simulation results demonstrated that the SAC-assisted controller improved the three-dimensional position tracking performance compared with the baseline controller without reinforcement learning assistance. Under the main OU-type stochastic wind disturbance condition, the SAC-assisted controller reduced the 3D position RMSE by 31.82%, 13.15%, 11.75%, and 26.29% in the hovering, line-segment, circular-path, and figure-eight-path scenarios, respectively. Compared with the DDPG- and TD3-assisted controllers, SAC provided the most consistent improvement among the tested RL-based auxiliary controllers under the evaluated trajectory and disturbance conditions. The additional reward-weight sensitivity analysis showed that the controller performance was more sensitive to the velocity-related weight λ_v than to the attitude-related weight λ_a, while the SAC-assisted controller maintained positive improvement over the No Agent baseline in most tested settings.

Additional simulation evaluations further supported the robustness and implementation feasibility of the proposed structure under the tested conditions. Under an unseen multi-harmonic sinusoidal wind disturbance, the SAC-assisted controller maintained positive 3D position RMSE improvements in all four scenarios, indicating that the learned residual compensation effect was not limited to the OU-type training disturbance. The thrust saturation analysis showed that SAC achieved the lowest auxiliary-thrust saturation rate among the tested RL algorithms and did not rely on sustained thrust saturation to improve tracking performance. In addition, the measured SAC inference time was sufficiently small relative to the 50 ms simulation control period in the tested CPU-based MATLAB environment.

The present study was conducted in a simulation environment. Future work will include experimental validation using a real quadrotor platform, implementation and timing evaluation on embedded flight hardware, and evaluation under more diverse disturbance conditions, including step-like gust disturbances. Further studies will also consider broader trajectory shapes, operating ranges, and additional training scenarios to examine the applicability of the residual policy under more realistic flight conditions.

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